



ASSESSMENT PLAN

For

Python Fundamental Course for Beginners

TGS Ref No: TGS-2019503161

**Conducted by
Tertiary Infotech Pte Ltd
UEN: 201200696W**

Version 4.0

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1. Version Control Record

Version	Effective Date	Changes	Who
1.0	18 Feb 2023	For Pilot Launch	Tertiary Infotech
1.1	23 Sep 2019	Added the following to the assessment plan • Assessment spec • Instructions to assessor and candidates • Evidence gathering plan Code of Practice to assessors	Tertiary Infotech
1.2	6 Nov 2019	Corrected some minor typo	Tertiary Infotech
1.3	11 Dec 2019	Corrected some minor typo	Tertiary Infotech
1.4	11 Dec 2019	Corrected some minor typo	Tertiary Infotech
1.5	12 Dec 2019	Corrected some minor typo	Tertiary Infotech
1.6	9 Mar 2020	Corrected some minor typo	Tertiary Infotech
1.7	1 Jun 2020	Modified the OQ	Tertiary Infotech
1.8	29 Feb 2024	Minor change in format	Tertiary Infotech
2.0	3 Jul 2024	Update on the format • Added the appeal procedure • Added evidence collection and submission	Tertiary Infotech
2.1	2 Jan 2025	• Changed “Developer” column To “Who” • Change everything to Tertiary Infotech instead of name	Tertiary Infotech
3.0	08 Jan 2025	Added Assessment Questions & Answers into Annex D	Tertiary Infotech
4.0	07 Apr 2025	Revised Assessment Plan format	Tertiary Infotech

2. Overview of Assessment

The purpose of this assessment is to evaluate the candidate's performance for **Python Fundamental Course for Beginners** against the following technical skills and competencies (TSC)

Skills Framework:

TSC Title: **Analytics and Computational Modelling**

TSC Code: **ICT-DIT-3001-1.1**

Assessment Methods

Learning Unit	Assessment Method(s)
LU1: Install Python and IDE	WA-SAQ
LU2: Understand and code data types	WA-SAQ, CS
LU3: Understand and code operators	WA-SAQ, CS
LU4: Understand and code control structures, loop and comprehension	WA-SAQ, CS
LU5: Understand and code functions	WA-SAQ
LU6: Install and use Python standard packages and third party package	WA-SAQ, OQ

Legend: **WA-SAQ**: Written Assessment (Short Answer Questions), **PP**: Practical Performance, **CS**: Case Study, **OQ**: Oral Questioning, **RP**: Role Play

Assessment Duration (per candidate)

Assessment Method(s)	Assessor to Candidate Ratio	Duration
WA-SAQ	1:1 (Min) 1:10 (Max)	60 mins
CS	1:1 (Min) 1:10 (Max)	70 mins
OQ	1:1 (Min) 1:1 (Max)	20 mins
Total time		150 mins (2.5 hrs)

Assessment Venue

Assessment Method(s)	Venue
WA-SAQ	Tertiary Infotech Pte Ltd or Client's Premise
CS	Tertiary Infotech Pte Ltd or Client's Premise
OQ	Tertiary Infotech Pte Ltd or Client's Premise

3. Matrix of Assessment Methods and Evidence Gathering Plan

3.1. Knowledge Assessment Matrix

Knowledge	WA-SAQ	CS	OQ
K1: Types of algorithms and advanced computational methods	√	-	-
K2: Range and application of various statistical algorithms	√	-	-
K3: Range and application of various types of data models	√	-	-
K4: Conduct statistical modelling of data to derive patterns / solutions	√	-	-
K5: Statistical modelling techniques	√	-	√
K6: Coding languages for Programming of Algorithms and Signals	√	-	√
K7: Potential reasons for unintended outcomes	√	-	√

3.2. Abilities Assessment Matrix

Ability	WA-SAQ	CS	OQ
A1: Identify appropriate statistical algorithms and data models to test hypotheses or theories	√	-	-
A2: Use appropriate analytics platforms and analytical tools given specific analytics and reporting requirements	√	√	-
A3: Utilise a range of statistical methods and analytics approaches to data	√	√	-
A4: Conduct statistical modelling of data to derive patterns / solutions	√	√	-
A5: Perform coding and configuration of software agents or programs based on a selected model or algorithm	√	√	-
A6: Conduct tests on the actions taken and outcomes to assess effectiveness of the model	√	√	-
A7: Diagnose unintended outcomes produced by analytical models	√	√	-
A8: Propose changes or updates to the model or algorithms applied	√	√	-
A9: Implement changes to the coding and configuration of software	√	√	-
A10: Draw relevant trends and insights from data analysis to support decisions	√	√	-

3.3. Evidence Gathering Plan

Assessment Methods	Evidence Gathering Plan
Written Assessment - Short Answer Questions Individual, Summative, Open book	Type of Evidence: Written responses to short-answer questions. Manner of Submission: Candidates will generate answers to the questions and submit the hardcopy to the assessor. Marking Process: Assessors will evaluate the written responses on the accuracy and comprehensiveness of the answers. Retention Period: 3 years.
Case Study Individual, Summative, Open book	Type of Evidence: • LO1: Candidates will submit a case study report detailing the installation process of Python and an IDE, including challenges faced. • LO2: Candidates will analyze a dataset, documenting their approach to selecting appropriate data types for various data points. • LO3: Candidates will present a case study on the use of operators in a real-world application, including code snippets. • LO4: Candidates will provide a case study on control structures and loops used in a specific project, detailing their effectiveness. • LO5: Candidates will document the design and implementation of functions in a case study format, including examples. • LO6: Candidates will analyze the impact of using standard and third-party packages in a case study. Manner of Submission: • Candidates will submit their case study reports electronically via the learning management system. Marking Process: • Clarity of Analysis. • Depth of Understanding. • Application of Concepts. Retention Period: All submitted case study reports will be retained for 3 years.
Oral Questioning Individual, Summative	Type of Evidence: • LO1: Candidates will explain the steps taken to install Python and an IDE during the oral questioning session. • LO2: Candidates will describe different data types and provide examples during the oral questioning. • LO3: Candidates will discuss the various operators in Python and their use cases during the oral questioning. • LO4: Candidates will explain control structures and loops, providing examples during the oral questioning. • LO5: Candidates will describe the purpose and structure of functions in Python during the oral questioning. • LO6: Candidates will discuss the use of standard and third-party packages in Python during the oral questioning. Manner of Submission: • Candidates will verbally respond to assessors during a structured questioning session.

	Marking Process: • Clarity of Explanation. • Depth of Knowledge. • Relevance of Examples. Retention Period: All oral questioning recordings and assessment notes will be retained for 2 years for compliance and auditing.
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4. Assessment Method Specifications

4.1. Assessment Specifications

4.1.1. Assessment Specification for Written Assessment - Short Answer Questions

Specifications	Written Assessment - Short Answer Questions
Abilities and Knowledge	K1, K2, K3, K4, K5, K6, K7, A1, A2, A3, A4, A5, A6, A7, A8, A9, A10
Duration	1 hr
Venue	Training room / designated assessment area
Set up	<u>Physical Assessment (Main)</u> Candidates are required to sit in a manner whereby there is sufficient separation so that they will not be able to interact during the assessment. <u>Virtual Assessment</u> <u>LMS</u> • Questions to be pre-loaded into LMS and opened only to the candidates and during the assessment time. • Alternatively, assessors will send the questions electronically to candidates only during the assessment time. <u>Virtual Environment</u> • Learners to ensure the camera is on all the time and showing their full face. • Learners to share their whole screen throughout the assessment period • Assessors to ensure the recording is started before the candidates and remained recordings for every breakout room throughout all the duration of the Assessment.
Pre-assessment instructions for assessors	• Put the candidate at ease • Explain clearly the purpose and context of the assessment: • Purpose of the assessment • Assessment duration • Assessment methods and tools • Explain the methods to be used, the standards and criteria for the assessment and the duration of the assessment. • Explain the procedures of the assessment: • Check whether candidate has any special requirements e.g. language, special needs, etc. • Ensure that the following documents are made available • Assessment Summary Record • Candidate's Appeal Form • Brief candidate on his/her rights and process of appeal • Identify any allowable adjustments that candidates may

	require should they have a special need In addition to the usual pre-assessment instructions, Include instructions on use of: ● Assessment Code of Practice
Pre-assessment instructions for candidates	● Be punctual ● Observe dress code: smart casual ● Request for assessor to repeat questions if question is not clear or if candidate has difficulty understanding assessor's accent ● To specify special needs, if any
Process of conducting assessment	● Verification of candidate's identity ● Ensure candidates are seated properly so that they will not interact with each other ● Encourage the candidate to seek clarification if any ● Keep in mind the time limit for the assessment
Managing limitations of evidence	● Use of Oral Clarification
Recording Assessment Results	● Record the results in the Assessment Summary Record ● Check that all record forms, timing, dates, names, NRIC, signatures are completed. ● Check that all Assessment Criteria are ticked competent (C) or not yet competent (NYC) in the specific questions. As long as there is a NYC, the candidate is considered not successful in the entire assessment. The candidate is required to re-take the entire assessment. ● Record observation of Competence where applicable in the 'Evidence of C and NYC must be recorded column ● Record reasons of Not Yet Competence where applicable in the 'Evidence of C and NYC must be recorded column ● An 'NYC' in any K or A will results in an 'NYC' in the unit. A candidate is only deemed competent when they achieved a 'C' for all the A and K in the unit.
Feedback	● Give clear and constructive feedback to candidate using appropriate language ● Seek clarification and agreement with the candidate on the outcome ● Provide guidance on overcoming gaps in competency ● Ask the candidates to sign the assessment record ● Record the feedback on the assessment record ● Sign off the assessment record ● Assessors should reiterate the appeal process (if the candidate is assessed as not yet competent)

4.1.2. Assessment Specification for Case Study

Specifications	Case Study
Abilities and Knowledge	A2, A3, A4, A5, A6, A7, A8, A9, A10
Duration	1.2 hrs
Venue	Training room / designated assessment area
Set up	<u>Physical Assessment (Main)</u> Candidates are required to sit in a manner whereby there is sufficient separation so that they will not be able to interact during the assessment. <u>Virtual Assessment</u> <u>LMS</u> • Questions to be pre-loaded into LMS and opened only to the candidates and during the assessment time. • Alternatively, assessors will send the questions electronically to candidates only during the assessment time. <u>Virtual Environment</u> • Learners to ensure the camera is on all the time and showing their full face. • Learners to share their whole screen throughout the assessment period • Assessors to ensure the recording is started before the candidates and remained recordings for every breakout room throughout all the duration of the Assessment.
Pre-assessment instructions for assessors	• Put the candidate at ease • Explain clearly the purpose and context of the assessment: • Purpose of the assessment • Assessment duration • Assessment methods and tools • Explain the methods to be used, the standards and criteria for the assessment and the duration of the assessment. • Explain the procedures of the assessment: • Check whether candidate has any special requirements e.g. language, special needs, etc. • Ensure that the following documents are made available • Assessment Summary Record • Candidate's Appeal Form • Brief candidate on his/her rights and process of appeal • Identify any allowable adjustments that candidates may require should they have a special need In addition to the usual pre-assessment instructions, Include instructions on use of:

	● Assessment Code of Practice
Pre-assessment instructions for candidates	● Be punctual ● Observe dress code: smart casual ● Request for assessor to repeat questions if question is not clear or if candidate has difficulty understanding assessor's accent ● To specify special needs, if any
Process of conducting assessment	● Verification of candidate's identity ● Ensure candidates are seated properly so that they will not interact with each other ● Encourage the candidate to seek clarification if any ● Keep in mind the time limit for the assessment
Managing limitations of evidence	● Use of Oral Clarification
Recording Assessment Results	● Record the results in the Assessment Summary Record ● Check that all record forms, timing, dates, names, NRIC, signatures are completed. ● Check that all Assessment Criteria are ticked competent (C) or not yet competent (NYC) in the specific questions. As long as there is a NYC, the candidate is considered not successful in the entire assessment. The candidate is required to re-take the entire assessment. ● Record observation of Competence where applicable in the 'Evidence of C and NYC must be recorded column ● Record reasons of Not Yet Competence where applicable in the 'Evidence of C and NYC must be recorded column ● An 'NYC' in any K or A will result in an 'NYC' in the unit. A candidate is only deemed competent when they achieved a 'C' for all the A and K in the unit.
Feedback	● Give clear and constructive feedback to candidate using appropriate language ● Seek clarification and agreement with the candidate on the outcome ● Provide guidance on overcoming gaps in competency ● Ask the candidates to sign the assessment record ● Record the feedback on the assessment record ● Sign off the assessment record ● Assessors should reiterate the appeal process (if the candidate is assessed as not yet competent)

4.1.3. Assessment Specification for Oral Questioning

Specifications	Oral Questioning
Abilities and Knowledge	K5, K6, K7
Duration	0.3 hrs
Venue	Training room / designated assessment area
Set up	<u>Physical Assessment (Main)</u> Candidates are required to sit in a manner whereby there is sufficient separation so that they will not be able to interact during the assessment. <u>Virtual Assessment</u> <u>LMS</u> • Questions to be pre-loaded into LMS and opened only to the candidates and during the assessment time. • Alternatively, assessors will send the questions electronically to candidates only during the assessment time. <u>Virtual Environment</u> • Learners to ensure the camera is on all the time and showing their full face. • Learners to share their whole screen throughout the assessment period • Assessors to ensure the recording is started before the candidates and remained recordings for every breakout room throughout all the duration of the Assessment.
Pre-assessment instructions for assessors	• Put the candidate at ease • Explain clearly the purpose and context of the assessment: • Purpose of the assessment • Assessment duration • Assessment methods and tools • Explain the methods to be used, the standards and criteria for the assessment and the duration of the assessment. • Explain the procedures of the assessment: • Check whether candidate has any special requirements e.g. language, special needs, etc. • Ensure that the following documents are made available • Assessment Summary Record • Candidate's Appeal Form • Brief candidate on his/her rights and process of appeal • Identify any allowable adjustments that candidates may require should they have a special need In addition to the usual pre-assessment instructions, Include instructions on use of:

	● Assessment Code of Practice
Pre-assessment instructions for candidates	● Be punctual ● Observe dress code: smart casual ● Request for assessor to repeat questions if question is not clear or if candidate has difficulty understanding assessor's accent ● To specify special needs, if any
Process of conducting assessment	● Verification of candidate's identity ● Ensure candidates are seated properly so that they will not interact with each other ● Encourage the candidate to seek clarification if any ● Keep in mind the time limit for the assessment
Managing limitations of evidence	● Use of Oral Clarification
Recording Assessment Results	● Record the results in the Assessment Summary Record ● Check that all record forms, timing, dates, names, NRIC, signatures are completed. ● Check that all Assessment Criteria are ticked competent (C) or not yet competent (NYC) in the specific questions. As long as there is a NYC, the candidate is considered not successful in the entire assessment. The candidate is required to re-take the entire assessment. ● Record observation of Competence where applicable in the 'Evidence of C and NYC must be recorded column ● Record reasons of Not Yet Competence where applicable in the 'Evidence of C and NYC must be recorded column ● An 'NYC' in any K or A will result in an 'NYC' in the unit. A candidate is only deemed competent when they achieved a 'C' for all the A and K in the unit.
Feedback	● Give clear and constructive feedback to candidate using appropriate language ● Seek clarification and agreement with the candidate on the outcome ● Provide guidance on overcoming gaps in competency ● Ask the candidates to sign the assessment record ● Record the feedback on the assessment record ● Sign off the assessment record ● Assessors should reiterate the appeal process (if the candidate is assessed as not yet competent)

5. Assessment Code of Practice

5.1 This code of practice provides:

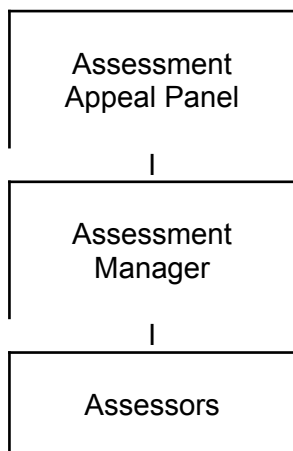
- Assessors with direction on the standard of practice.
- Candidates with assurance of the standards of practice expected of assessors.
- Employers with assurance of the standards maintained by Tertiary Infotech Pte Ltd in the conduct of assessment.

5.2 The following code of practice is based on the international code of ethics and practice (The National Council for Measurement in Education [NCME]).

1. The differing needs and requirements of the person(s) being assessed, the local enterprise(s) and/or industry are identified and handled with sensitivity.
2. Potential forms of conflict of interest in the assessment process and/or outcomes are identified and appropriate referrals are made, if necessary.
3. All forms of harassment are avoided throughout the planning, conducting, reviewing and reporting of the assessment outcomes.
4. The rights of the candidate(s) are protected during and after the assessment.
5. Personal or interpersonal factors that are not relevant to the assessment of competency must not influence the assessment outcomes.
6. The candidate is made aware of rights and processes of appeal.
7. Evidence that is gathered during the assessment is verified for validity, reliability, authenticity, sufficiency and currency.
8. Assessment decisions are based on available evidence that can be produced and verified by another assessor.
9. Assessments are conducted within the boundaries of the assessment system policies and procedures.
10. Formal agreement is obtained from both the candidate(s) and the assessor that the assessment was carried out in accordance with agreed procedures.
11. Assessment tools, systems, and procedures are consistent with equal opportunity legislation.
12. The candidate is informed of all assessment reporting processes prior to the Assessment.
13. The candidate is informed of all known potential consequences of decisions arising from an assessment, prior to the assessment.
14. Confidentiality is maintained regarding assessment results.
15. Results are only released with the written permission of the candidate(s).
16. The assessment results are used consistently with the purposes explained to the candidate.
17. Self-assessments are periodically conducted to compare current competencies against the required competencies.
18. Professional development opportunities are identified and sought.
19. Opportunities for networking amongst assessors are created and maintained.
20. Opportunities are created for technical assistance in planning, conducting and reviewing assessment procedures and outcomes.

6. Assessment Management Organisation

Assessment Appeal Panel



6.1. Assessment Appeal Panel

- The Assessment Appeal Panel is responsible for reviewing and giving a decision on appeals against a Not-Yet-Competent award
- The panel is comprised of a Management Representative, the Assessment Manager and one Assessor who is independent of the case in question.

6.2. Assessment Manager

The Assessment Manager is responsible:

- For the administration and conduct of assessment
- Monitoring and continuous improvement of the assessment process and tools
- For the continuous professional development of the pool of assessors
- Managing the process and outcome of the appeal cases.

6.3. Assessor

- The Assessor is responsible for conducting the assessment according to the Assessment Plan.

6.4. Appeal Procedures

The candidate has the right to challenge the assessment decision made by the Assessor. When giving feedback to the candidate about the assessment, the Assessor must ask the candidate if they agree with the outcome.

If the candidate agrees with the outcome, the Assessor and the candidate must sign the Assessment Outcome record. If the candidate does not agree with the outcome, the candidate should not sign the Assessment Outcome record.

If the candidate intends to appeal against the decision made, he/she should discuss the matter with the Assessor. If he/she is still not satisfied with the decision, the candidate must notify the Assessor of the intention to appeal. The Assessor will enter the intention in the Feedback section of the Assessment Outcome record.

The Assessor will notify the training provider of the candidate's intention to lodge an appeal.

The Assessment Manager will collect information from the Assessor. A separate session will be arranged for the candidate and an independent Assessor appointed by the training provider. An outcome agreed by all will conclude the appeal procedure.

A record of the appeal and any subsequent actions and findings will be submitted to SSG for record purposes.

7. Instructions to Assessor

A. Prior to the Assessment, I...	Tick	Remarks
1. ensured that the candidate is informed regarding the venue and schedule of assessment		
2. reviewed the current copies of the performance statement to be assessed, the assessment and evidence gathering plan, the assessment checklist and the organisations standard operating procedures (SOP)		
3. reviewed the performance statement and evidence plan to ensure that I clearly understood the instructions and the requirements of the assessment process.		
4. identified and accommodated any special needs candidate		
5. checked the logistics set up and resources for the assessment.		
B. During the Assessment, I:		
1. introduced myself and confirmed the identities of the candidates.		
2. put the candidates at ease by being friendly and helpful.		
3. explained to the candidates the purpose, context and benefits of the assessment.		
4. ensured the candidates understood the process of the assessment.		
5. gave the candidates an overview of performance statement to be assessed.		
6. explained the appeal procedure and the results reporting procedure.		
7. encouraged candidates to seek clarifications if in doubt.		
8. asked candidates for feedback on the assessment.		
9. explained appropriate legal, safety and ethical issues		
C. After the Assessment, I:		
1. ensured that the candidate was given constructive feedback.		
2. completed and signed the Individual Assessment Summary Record and the Assessment Record – Class Summary		
3. thanked the candidate for participating in the assessment.		

8. Instructions to Candidate

8.1. Pre-Assessment Information for Candidate\

Assessors are required to provide candidates with relevant pre-assessment information which includes, but is not limited

- 1) This assessment of a range of assessment methods, including Written Assignment (Q&A), Written Assignment (Case Study) and Oral Questioning. The conduct of the assessment may be one-on-one, and, in a group, context as outlined in the Assessment Specifications.
- 2) For these competency elements and the required performance criteria (including range and context) will be assessed. Refer to the Assessment Specifications and 'Instructions for Candidates' for a detailed coverage of the assessment requirements.
- 3) You have a right of appeal to assessment decisions in accordance with Tertiary Infotech Pte Ltd appeal procedures
- 4) The conditions for granting a deferred assessment are:
 - Medical grounds – a medical report or certificate from registered medical practitioners
 - Unexpected and exceptional circumstances which may include (but are not limited to):
 - accidents (sporting, motor vehicle, etc.) where an injury is sustained bereavement

Unexpected and exceptional circumstances **do not** include cases where candidates have mistaken the day, time or venue of assessment.

9. Assessment Records

9.1. Assessment Record for Written Assessment - Short Answer Questions

Learning Outcomes	TSC Abilities and Knowledge	Please Tick		Remarks
		C	NYC	
LO1 - Learners will be able to install Python and IDE	K1: Types of algorithms and advanced computational methods A1: Identify appropriate statistical algorithms and data models to test hypotheses or theories			
LO2 - Learners will be able to understand and code data types	K2: Range and application of various statistical algorithms K3: Range and application of various types of data models A2: Use appropriate analytics platforms and analytical tools given specific analytics and reporting requirements A3: Utilise a range of statistical methods and analytics approaches to data A4: Conduct statistical modelling of data to derive patterns / solutions			
LO3 - Learners will be able to understand and code operators	K4: Conduct statistical modelling of data to derive patterns / solutions K5: Statistical modelling techniques K6: Coding languages for programming of algorithms and signals A5: Perform coding and configuration of software agents or programs based on a selected model or algorithm A6: Conduct tests on the actions taken and outcomes to assess effectiveness of the model			

LO4 - Learners will be able to understand and code control structures, loop and comprehension.	K5: Statistical modelling techniques K6: Coding languages for Programming of Algorithms and Signals A4: Perform data analysis using basic statistical methods and techniques, to determine the relationship between variables A5: Perform coding and configuration of software agents or programs based on a selected model or algorithm A6: Conduct tests on the actions taken and outcomes to assess effectiveness of the model			
LO5 - Learners will be able to understand and code functions	K5: Statistical modelling techniques K6: Coding languages for programming of algorithms and signals A7: Diagnose unintended outcomes produced by analytical models A9: Implement changes to the coding and configuration of software			
LO6 - Learners will be able to install and use Python standard packages and third party package	K7: Potential reasons for unintended outcomes A8: Propose changes or updates to the model or algorithms applied A10: Draw relevant trends and insights from data analysis to support decisions			

C: Competent; NYC: Not Yet Competent

9.2. Assessment Record for Oral Questioning

Learning Outcomes	TSC Abilities and Knowledge	Please Tick		Remarks
		C	NYC	
LO1 - Learners will be able to install Python and IDE	-			
LO2 - Learners will be able to understand and code data types	-			
LO3 - Learners will be able to understand and code operators	-			
LO4 - Learners will be able to understand and code control structures, loop and comprehension.	-			
LO5 - Learners will be able to understand and code functions	K5: Statistical modelling techniques K6: Coding languages for programming of algorithms and signals			
LO6 - Learners will be able to install and use Python standard packages and third party package	K7: Potential reasons for unintended outcomes			

C: Competent; NYC: Not Yet Competent

9.3. Assessment Record for Case Study

Enabling Learning Outcomes	TSC Abilities and Knowledge	Please Tick		Remarks
		C	NYC	
LO1 - Learners will be able to install Python and IDE	-			
LO2 - Learners will be able to understand and code data types	A2: Use appropriate analytics platforms and analytical tools given specific analytics and reporting requirements A3: Utilise a range of statistical methods and analytics approaches to data A4: Conduct statistical modelling of data to derive patterns / solutions			
LO3 - Learners will be able to understand and code operators	A5: Perform coding and configuration of software agents or programs based on a selected model or algorithm A6: Conduct tests on the actions taken and outcomes to assess effectiveness of the model			
LO4 - Learners will be able to understand and code control structures, loop and comprehension.	A4: Perform data analysis using basic statistical methods and techniques, to determine the relationship between variables A5: Perform coding and configuration of software agents or programs based on a selected model or algorithm A6: Conduct tests on the actions taken and outcomes to assess effectiveness of the model			
LO5 - Learners will be able to understand and code functions	A7: Diagnose unintended outcomes produced by analytical models A9: Implement changes to the coding and configuration of software			

LO6 - Learners will be able to install and use Python standard packages and third party package	A8: Propose changes or updates to the model or algorithms applied A10: Draw relevant trends and insights from data analysis to support decisions			
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C: Competent; NYC: Not Yet Competent

9.4. Assessment Summary Record

Learning Outcomes	WA-SAQ	CS	OQ	Competence (Please Tick) C or NYC
LO1 - Learners will be able to install Python and IDE	√	-	-	
LO2 - Learners will be able to understand and code data types	√	√	-	
LO3 - Learners will be able to understand and code operators	√	√	-	
LO4 - Learners will be able to understand and code control structures, loop and comprehension.	√	√	-	
LO5 - Learners will be able to understand and code functions	√	√	√	
LO6 - Learners will be able to install and use Python standard packages and third party package	√	√	√	

Legend: **WA-SAQ**: Written Assessment (Short Answer Questions), **PP**: Practical Performance, **CS**: Case Study, **OQ**: Oral Questioning, **RP**: Role Play, **NA**: Not Applicable, **C**: Competent, **NYC**: Not Yet Competent

This candidate has been assessed as

☐

COMPETENT

☐

NOT YET COMPETENT

Candidate Name (As in NRIC)		Assessor Name	
NRIC (Last 3 digits & alphabet)		NRIC (Last 3 digits & alphabet)	
Candidate Signature		Assessor Signature	
Date:		Date:	
<i>By signing, the candidate is agreeing to accept the assessment outcome</i>		<i>By signing, the assessor is agreeing to have duly assessed the performance criteria and underpinning knowledge as required in the competence units</i>	

Feedback on outcome by Assessor/ Feedback by candidate:

{Feedback on the overall performance or in the case of NYC; any area of skills gap and improvement needed}

10. Annexes

10.1. Assessment Questions and Answers

10.1.1. Annex A: Assessment Questions for WA-SAQ

10.1.2. Annex B: Suggested Answer to Assessment Questions for WA-SAQ

10.1.3. Annex C: Assessment Questions for CS

10.1.4. Annex D: Suggested Answer to Assessment Questions for CS

10.1.5. Annex E: Assessment Questions for OQ

10.1.6. Annex F: Suggested Answer to Assessment Questions for OQ

Annex A: QUESTION PAPER OF WA (SAQ) ASSESSMENT

Python Fundamental Course for Beginners

Written Assessment (SAQ)

A: Trainee Information:

Trainee Name (as Per NRIC): _____

Last 3 digits and alphabet of NRIC/FIN: _____

Date: _____

B: Instruction to Candidates

1. This is an individual exercise
2. This is an open book assessment
3. A total of 1 hr is given to complete the written assessment
4. Submit your answers on the document provided

C: Short Answer Questions

Question 1: Imagine you're a Python programmer working on a calculator app. You're developing a feature for your app that involves performing complex arithmetic operations in a specific order. To optimize the code for speed and readability, you decide to leverage Python's tuple unpacking.

Analyze the code snippet from the calculator app below:

```
a, b = 4, 6  
a, b = b, a+b
```

What is the final value of *b*? (K1)

Question 2: You're tasked with analyzing product sales trends to identify popular product categories and inform inventory management decisions. To accurately analyze the data, you need to ensure that product names are consistent across the dataset. This involves handling variations in capitalization, punctuation, and extra spaces, which requires cleaning and standardizing the product names before proceeding with your analysis.

Analyze the code snippet from your data cleaning below:

```
a = 'Hello'  
b = a.capitalize().lower()
```

What is the final value of *b*? (K2)

Question 3: You're developing a Python script to analyze numerical data related to product sales over a specific time period. You need to create a sequence of numbers representing certain data points and convert it into a list for further processing.

Study the following Python code snippet:

```
a = range(6, 3, -2)  
b = list(a)
```

What is the final value of *b*? (K3)

Question 4: Imagine you're building a simple color-coded system to categorize different types of items based on their numerical values. You're testing a preliminary logic to determine which color should be assigned to a given value.

Study the following Python code snippet:

```
a = 1
b = 2
if not (b == 3):
    print("red")
else:
    print("blue")
```

What is printed to the console? (K4)

Question 5: You're working on a Python script to model a population growth simulation. You're using variables to represent different segments of the population and defining rules for how these segments interact and grow over time. The code snippet you're analyzing represents a simplified version of this model where you're tracking the population of three different age groups.

Study the following Python code snippet:

```
a, b, c = 1, 1, 1
while b < 10:
    a, b, c = b, c, a + b + c
```

What is the final value of *c*? (K5)

Question 6: You're working on a data analysis project involving product pricing. You need to apply a specific transformation to a list of base prices and store the results in a new list for further analysis.

Study the following Python code snippet:

```
a = [2, 3, 4]
f = lambda x: 3 * x
b = map(f, a)
c = list(b)
```

What is the final value of *c*? (K6)

Question 7: Imagine you're a data analyst working for a marketing firm. Your team has collected a large dataset on customer demographics, product preferences, and purchase history. You've cleaned and processed the data, and now you need to create compelling visualizations to present insights to the marketing and sales teams.

Which third-party Python package can you use for data visualization? (K7)

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Grade: _____(C / NYC)

Assessor Name: _____

Assessor NRIC: _____

Date: _____

Signature: _____

Note: Please create a subfolder named YYYY-MM-DD/DD_<<Trainer Name>> and upload all the learner's assessments to the respective Google Drive.

Please email erra@tertiaryinfotech.com after you have uploaded the assessment.

Annex B:

SUGGESTED ANSWER TO WA (SAQ) ASSESSMENT QUESTIONS

Python Fundamental Course for Beginners

Answers to Written Assessment (SAQ)

C: Short Answer Questions

Question 1:

Imagine you're a Python programmer working on a calculator app. You're developing a feature for your app that involves performing complex arithmetic operations in a specific order. To optimize the code for speed and readability, you decide to leverage Python's tuple unpacking.

Analyze the code snippet from the calculator app below:

```
a, b = 4, 6  
a, b = b, a+b
```

What is the final value of *b*? (K1)

Suggestive answers (not exhaustive):

- 10

Question 2:

You're tasked with analyzing product sales trends to identify popular product categories and inform inventory management decisions. To accurately analyze the data, you need to ensure that product names are consistent across the dataset. This involves handling variations in capitalization, punctuation, and extra spaces, which requires cleaning and standardizing the product names before proceeding with your analysis.

Analyze the code snippet from your data cleaning below:

```
a = 'Hello'  
b = a.capitalize().lower()
```

What is the final value of *b*? (K2)

Suggestive answers (not exhaustive):

- hello

Question 3:

You're developing a Python script to analyze numerical data related to product sales over a specific time period. You need to create a sequence of numbers representing certain data points and convert it into a list for further processing.

Study the following Python code snippet:

```
a = range(6, 3, -2)
b = list(a)
```

What is the final value of *b*? (K3)

Suggestive answers (not exhaustive):

- [6, 4]

Question 4:

Imagine you're building a simple color-coded system to categorize different types of items based on their numerical values. You're testing a preliminary logic to determine which color should be assigned to a given value.

Study the following Python code snippet:

```
a = 1
b = 2
if not (b == 3):
    print("red")
else:
    print("blue")
```

What is printed to the console? (K4)

Suggestive answers (not exhaustive):

- red

Question 5:

You're working on a Python script to model a population growth simulation. You're using variables to represent different segments of the population and defining rules for how these segments interact and grow over time. The code snippet you're analyzing represents a simplified version of this model where you're tracking the population of three different age groups.

Study the following Python code snippet:

```
a, b, c = 1, 1, 1
while b < 10:
    a, b, c = b, c, a + b + c
```

What is the final value of *c*? (K5)

Suggestive answers (not exhaustive):

Question 6:

You're working on a data analysis project involving product pricing. You need to apply a specific transformation to a list of base prices and store the results in a new list for further analysis.

Study the following Python code snippet:

```
a = [2, 3, 4]
f = lambda x: 3 * x
b = map(f, a)
c = list(b)
```

What is the final value of *c*? (K6)

Suggestive answers (not exhaustive):

- [6, 9, 12]

Question 7:

Imagine you're a data analyst working for a marketing firm. Your team has collected a large dataset on customer demographics, product preferences, and purchase history. You've cleaned and processed the data, and now you need to create compelling visualizations to present insights to the marketing and sales teams.

Which third-party Python package can you use for data visualization? (K7)

Suggestive answers (not exhaustive):

- Matplotlib

Annex C: QUESTION PAPER OF CS ASSESSMENT

Python Fundamental Course for Beginners

Case Study Assessment

A: Trainee Information:

Trainee Name (as Per NRIC): _____

Last 3 digits and alphabet of NRIC/FIN: _____

Date: _____

B: Assessment Instruction

This is a Written Assessment (Case Study)

Duration: 70 mins

1. The assessor will pass the case study in hardcopy to you.
2. You need to code a Python program in a new Google Colab project.
3. This is an open book exam that must be completed individually.

Submission Procedure:

1. At the Google Colab, Goto File -> Download as .ipynb
2. Rename your file as this format TGS-2019503161_JohnTan_XXXX567A
3. Email the Jupyter file to the assessor

C. Case Study:

The Central Provident Fund (CPF) Board has engaged you to create a CPF contribution calculator. Using the user's age and their salary, the program should display the employer and employee CPF contributions to the user.

The CPF contribution rates are shown below:

Employee's age (years)	Contribution Rates from 1 Jan 2016 (for monthly wages ≥ \$750)		
	By Employer (% of wage)	By Employee (% of wage)	Total (% of wage)
55 and below	17	20	37
Above 55 to 60	13	13	26
Above 60 to 65	9	7.5	16.5
Above 65	7.5	5	12.5

Example prompts when running the application:

What is your age? 42

What is your monthly salary? 5000

Employer CPF contribution is: \$850.00

Employee CPF contribution is: \$1000.00

Question 1: Create a function called `get_user_age()` that prompts the user to enter their age for the CPF calculations.

- If the user is younger than 16 years old, the application should display the error message “Minimum age for CPF contribution is 16. Please try again.” and re-prompt the user.
- If the user is at least 16 years old, the function should return the user’s age as an integer

Show your Python code below: (A1, A2)

Question 2: Create a function called `get_user_salary()` that prompts the user to enter their monthly salary for the CPF calculations.

- If the user inputs a negative number, the application should display the error message “Salary cannot be negative. Please try again.” and re-prompt the user.
- If the user inputs a valid number, the function should return the user’s salary as a float.

Show your Python code below: (A3, A4)

Question 3: While testing out your code, you realized that the user can input invalid values for their age and salary (e.g., “abc”). To this, you decided to add try-except blocks to your functions `get_user_age()` and `get_user_salary()` to make them more robust to handle such edge cases.

Show your Python code below: (A5, A6)

Question 4: Create a function called `get_cpf_contributions()` that takes in two parameters – the employee’s age and salary – and computes the employer and employee CPF contributions.

Refer to the CPF contribution rates table above and show your Python code below: (A7, A8)

Question 5: Combining all your code from Questions 1 to 4,

- Prompt the user for their age and salary + handle invalid values
- Calculate the employer and employee CPF contributions
- Display the results to the user (monetary values should be formatted to 2 decimal places)

Show your Python code below: (A9, A10)

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Assessor NRIC: _____

Date: _____

Signature: _____

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Annex D:

SUGGESTED ANSWER TO CS ASSESSMENT QUESTIONS

Python Fundamental Course for Beginners

Answers to Written Assessment (Case Study)

C. Case Study:

Question 1:

Create a function called `get_user_age()` that prompts the user to enter their age for the CPF calculations.

- If the user is younger than 16 years old, the application should display the error message "Minimum age for CPF contribution is 16. Please try again." and re-prompt the user.
- If the user is at least 16 years old, the function should return the user's age as an integer

Show your Python code below: (A1, A2)

Suggested answers (non-exhaustive):

```
def get_user_age():
    """
    Prompts the user for their age.
    """
    while True:
        age = int(input("What is your age? "))
        if age < 16:
            print("Minimum age for CPF contribution is 16. Please try again.")
        else:
            return age
```

Question 2:

Create a function called `get_user_salary()` that prompts the user to enter their monthly salary for the CPF calculations.

- If the user inputs a negative number, the application should display the error message "Salary cannot be negative. Please try again." and re-prompt the user.
- If the user inputs a valid number, the function should return the user's salary as a float.

Show your Python code below: (A3, A4)

Suggested answers (non-exhaustive):

```
def get_user_salary():
    """
    Prompts the user for their salary.
    """
    while True:
        salary = float(input("What is your monthly salary? "))
        if salary < 0:
```

```
print("Salary cannot be negative. Please try again.")
else:
    return salary
```

Question 3:

While testing out your code, you realized that the user can input invalid values for their age and salary (e.g., "abc"). To this, you decided to add try-except blocks to your functions `get_user_age()` and `get_user_salary()` to make them more robust to handle such edge cases.

Show your Python code below: (A5, A6)

Suggested answers (non-exhaustive):

```
def get_user_age2():
    """
    Prompts the user for their age.
    """
    while True:
        try:
            age = int(input("What is your age? "))
            if age < 16:
                print("Minimum age for CPF contribution is 16. Please try again.")
            else:
                return age
        except ValueError:
            print("Invalid input. Please enter a valid age (integer)")

def get_user_salary2():
    """
    Prompts the user for their salary.
    """
    while True:
        try:
            salary = float(input("What is your monthly salary? "))
            if salary < 0:
                print("Salary cannot be negative. Please try again.")
            else:
                return salary
        except ValueError:
            print("Invalid input. Please enter a valid salary (decimal number)")
```

Question 4:

Create a function called `get_cpf_contributions()` that takes in two parameters – the employee's age and salary – and computes the employer and employee CPF contributions.

Refer to the CPF contribution rates table above and show your Python code below: (A7, A8)

Suggested answers (non-exhaustive):

```
def calculate_cpf_contribution(age: int, salary: float):
    """
    Calculates the employer and employee CPF contributions based on age and salary.

    Args:
        age: The employee's age
        salary: The employee's monthly salary

    Returns:
        A tuple containing `employer_contribution` and `employee_contribution` in that order.
    """

    # if monthly wages is less than $750, there is no CPF
    if salary < 750:
        return (0, 0)

    employer_percentage = -1
    employee_percentage = -1

    if age <= 55:
        employer_percentage = 0.17
        employee_percentage = 0.20
    elif 55 < age <= 60:
        employer_percentage = 0.13
        employee_percentage = 0.13
    elif 60 < age <= 65:
        employer_percentage = 0.09
        employee_percentage = 0.075
    elif age > 65:
        employer_percentage = 0.075
        employee_percentage = 0.05
    else:
        raise Exception("Unknown age")

    employer_contribution = salary * employer_percentage
    employee_contribution = salary * employee_percentage

    return (employer_contribution, employee_contribution)
```

Question 5:

Combining all your code from Questions 1 to 4,

- Prompt the user for their age and salary + handle invalid values
- Calculate the employer and employee CPF contributions
- Display the results to the user (monetary values should be formatted to 2 decimal places)

Show your Python code below: (A9, A10)

Suggested answers (non-exhaustive):

```
def main():
    age = get_user_age2()
    salary = get_user_salary2()

    employer_contribution, employee_contribution = calculate_cpf_contribution(
        age, salary
    )

    print(f"Employer CPF contribution is: ${employer_contribution:.2f}")
    print(f"Employee CPF contribution is: ${employee_contribution:.2f}")

if __name__ == "__main__":
    main()
```

Annex E: SUGGESTED ANSWER TO Oral Questioning ASSESSMENT QUESTIONS

Python Fundamental Course for Beginners

Answers to Oral Questioning (OQ)

Question 1: What is the use of function? (K5)

Suggestive answers (not exhaustive):

- Reuse the code snippet
- The codes are more modular
- Easier to maintain the codes

Question 2: What is the difference between normal function and lambda function? (K6)

Suggestive answers (not exhaustive):

- Unlike normal function, lambda function does not need a function name
- Unlike normal function, lambda function can be assigned to variable
- Unlike normal function, lambda function can be the input of another function
- Unlike normal function, there is no return statement
- Unlike normal function, lambda function can be the input of another function

Question 3: What is the reason if you encounter an error “name xxx is not defined? Eg: “name np is not defined” (K7)

Suggestive answers (not exhaustive):

- Did not install the third-party package
- Did not import the third-party package
- Did not define the alias eg import numpy as np
- Typo error